

A NOTE ON

A toy model of credit frictions

Content of the Handout

1	Outline.....	p. 1
2	The model	p. 1

1. Outline

2. The model

Let us now suppose that production is endogenous in the same spirit as ?. The welfare index of the representative household reads as follows:

$$\mathcal{W}_t = \mathbb{E}_t \sum_{\tau=0}^{+\infty} \theta_{t+\tau} \mathcal{U}(C_{jt+\tau}) \quad (1)$$

Note in this specific case here, we deal with an unit root component. To takeaway the unit root in the data generating process of consumption, we assume as ? that the stochastic discount factor is time varying:

$$\theta_t = \beta \exp(-\chi [C_t - \bar{C}] / \bar{C}), \quad (2)$$

where $\bar{C}$ is the steady state consumption and β the discount factor. Note that θ_t is economy-wide, so the agent j does not optimize over C_t .

The households faces a (intertemporal) budget constraint:

$$C_{jt} + I_{jt} = Y_{jt} \quad (3)$$

The total income Y_{jt} is spent on consumption goods C_{jt} and investment goods I_{jt} . An intertemporal budget constraint shows what consumption options are available between the present and future. The constraint can be used to determine the future value of present income or the present value of future income. This constraint binds each period, *i.e.* households expenditures equal her income for any value

of t . Here, the budget constraint can also be interpreted as the definition of the GDP using expenditure approach (neglecting both stocks and the trade balance).

The law of motion of capital reads as in the ?'s model:

$$I_{jt} = K_{jt} - (1 - \delta) K_{jt-1} \quad (4)$$

Investment is determined by the gap between current capital K_{jt} and depreciated capital $(1 - \delta) K_{jt-1}$. In the latter expression, the parameter $0 \leq \delta \leq 1$ denotes the depreciation rate of capital.

The standard Cobb-Douglas technology is determined by:

$$Y_{jt} = A_t K_{jt-1}^\alpha \quad (5)$$

where $0 < \alpha \leq 1$ is the share of capital involved in the production process of new goods Y_{jt} while A_t is a productivity shock which temporary affects the society's ability to produce goods, it affects all the households in our economy in the same way. The production combines technology and capital K_{jt-1} bought in $t - 1$ as capital requires one period to be settled (the so called time-to-build in the RBC literature). Concerning the productivity shock, it is assumed to follow a simple AR(1) process given by:

$$\log(A_t) = \rho \log(A_{t-1}) + \eta_t, \quad (6)$$